

**Title: Formula 1 Racing Car Components Reliability
Assessment and Monitoring System (FOCRAMS)**

Name: Yong Rui Qian

Student ID: 2022080054

Email: rqyong@gmail.com

Research Gap and Background

Despite advances in sensor fusion and predictive maintenance in various industries, a dedicated solution tailored to the extreme, high-stress environment of Formula 1 racing remains elusive. F1 cars operate under intense conditions—rapid acceleration, high temperatures, and significant vibrations—that demand real-time, predictive monitoring of component integrity. Current systems mainly focus on performance optimization or generic health monitoring, leaving a gap in integrating multi-modal sensor data for reliability assessment and proactive maintenance. In F1, even the slightest fault can result in catastrophic failures, race retirements, or costly downtime. With hundreds of sensors generating vast amounts of data during races, there is a critical need for a system that not only leverages this data for immediate performance insights but also forecasts potential degradation. This combined challenge of ensuring both optimal performance and preemptive fault detection underscores the necessity for a specialized, sensor-based monitoring solution in Formula 1.

Analysis of Domestic and International Research Status

I. Multi-Modal Sensor Fusion for Vehicle Health Monitoring

Modern Formula One and high-performance vehicles embed hundreds of sensors (vibration, temperature, pressure, strain, etc.) across engine, drivetrain and chassis. These multi-modal measurements must be fused to accurately detect faults. In general, sensor fusion algorithms combine data from diverse sensors to compensate for individual weaknesses and improve. For example, one study uses heterogeneous feature-level fusion of accelerometer (vibration) and electrical signals to detect faults more accurately than single-sensor models. Likewise, a Chinese patent (CN114821856B) by Dalian University of Technology describe adding “*quick-plug*” auxiliary sensors alongside a vehicle’s original sensors and electric control unit (ECU) to collect complementary data (e.g. extra thermal, vibration, pressure readings) and feed them into an AI-based diagnosis module.

II. Edge Computing & Real-Time Telemetry

To meet low-latency needs (<10 ms) and reduce data packet loss, FOCRAMS will use edge computing. Modern edge architectures place processing close to the sensors (“at the edge”) so that raw data need not travel to a central server. Edge devices preprocess and analyze data locally, reducing transmission and response time. For example, Kociszewski et al. designed an engine-vibration monitoring system where an accelerometer feeds data into a microcontroller for on-board Fast Fourier Transform (FFT) processing and Message Queue Telemetry Transport (MQTT) streaming. This approach “enables more efficient scaling” and reduces latency since only high-level features (or alerts) are sent off-board. Recent IoT studies confirm that TinyML models can run on resource-constrained edge hardware to detect motor faults with ~300 ms total latency and >96% accuracy. From a hardware standpoint, industrial automotive-edge computers now support 1 Gbps links and AI acceleration. For

example, a rugged NVIDIA Jetson-based unit supports four Single-Pair Ethernet (BroadR-Reach) ports (1000BASE-T1), each with its own Network Interface Card (NIC) for real-time sensor streams. This lets multiple high-speed channels (cameras, Inertial Measurement Units (IMUs), F1 sensors, etc.) connect with millisecond determinism.

III. High-Speed Vibration & Thermal Sensing

Vibration sensing: Bosch's Microelectromechanical Systems (MEMS) 3-axis accelerometers explicitly target "Vibration Monitoring" with 10 kHz bandwidth for predictive maintenance. Such sensors (often with Serial Peripheral Interface (SPI) or Inter-Integrated Circuit (I²C) output) can sample axes at $\geq 10,000$ samples/s, meeting our 10 kHz requirement. Data from these sensors will be sampled by a fast Analog-Digital Converter (ADC)/processor (e.g. 16–24 bit at 20 kHz) to detect bearing fault frequencies or gear mesh tones well beyond 5 kHz. These signals are then filtered/extracted on the edge unit to reduce raw data flow, sending only summaries or anomaly flags.

Thermal sensing: High-precision temperature sensors (thermistors or digital ICs) are required for oil, coolant, brake and bearing temperatures. Modern digital temp sensors routinely achieve ± 0.5 °C accuracy (16-bit converters) at <100 ms update rates. For example, the Bosch site notes "10 kHz bandwidth" only for vibration, but F1 teams use infrared and thermocouple sensors for engine surface and exhaust temps as well.

IV. Fluid Diagnostics & Sensor Technologies

In high-performance engines, multivariate fluid-condition sensors are used to track oil and brake-fluid health in real time. Modern systems use vibratory/MEMS viscometers (e.g. tuning-fork or capacitive MEMS devices) to measure oil viscosity. Simultaneously, moisture and particle counters (e.g. optical particle sensors, capacitive water sensors) monitor contamination. For example, a specialized F1 fluid-monitoring rig uses dozens of sensors (temperature, flow rate, pressure, particle count, humidity, water content, etc.) to give a comprehensive picture of fluid cleanliness. By fusing these signals, advanced diagnostics can deduce oil aging or brake-fluid degradation.

A recent patent (US2023/0211766A1 from GM Global Technology Operations LLC) even combines level, electrical conductivity (concentration), and temperature sensors in a brake reservoir to compute remaining fluid life, water content and acid levels. Similarly, general "oil health" patents note that fluid-property sensors (resonant forks, capacitive/dielectric probes, ultrasound, etc.) can feed algorithms to estimate additives depletion (TAN/TBN) and oxidation. Real-time IoT systems then use these measures to predict maintenance needs.

V. Brake-System Monitoring

Braking systems employ a variety of sensors to track component stress. High-precision temperature sensors (thermocouples or infrared pyrometers) monitor disc and pad heat in real time, since F1 brake temperatures can exceed 1000 °C. Pressure transducers in the hydraulic lines measure brake line pressure for every channel. Pad-wear sensors (traditional wired

contacts or inductive loops) signal pad replacement needs. Crucially, modern strain/force sensors quantify mechanical load: for example, an European Patent (EP3983268B1) embeds fiber-optic Bragg-grating sensors in a brake pad to measure both mechanical strain and temperature, allowing direct computation of braking force. Likewise, specialized brake-disc assemblies can house sensors and wireless transmitters: earlier patents describe placing sensors in or near the disk and using transponders to send temperature/strain data to the ECU. All these signals are logged per-brake to diagnose overheating, detect impending failure, and optimize brake balance.

Content of Research

I. Systems Indices

No.	Aspect	Index	Rationale
1.	Weight Constraint	Total weight of all sensors, edge processing units, and communications hardware must not exceed 400 g.	To avoid negatively impacting the car's handling and weight distribution hence hindering the car's performance.
2.	Latency Requirement	End-to-end data acquisition, processing, and transmission delay must be < 10 ms.	Ensures real-time feedback for control decisions, including safety-critical alerts.
3.	Thermal and Vibration Robustness	A. Sensor modules must operate reliably in ambient temperatures of -40 to +150°C, and local hotspots up to 250°C. B. All components must meet ISO 16750-3 standards for vibration resistance (>10 G RMS in 3 axes).	F1 components are exposed to severe thermal and mechanical stress.
4.	Data Fidelity	A. Vibration sensors must support ≥ 10 kHz sampling frequency, 16-bit resolution. B. Temperature sensors must have $\pm 0.5^\circ\text{C}$ precision and 100 Hz update rate. C. Fluid-quality sensors must detect viscosity changes within $\pm 5\%$, and water/contaminant content within 100 ppm.	Ensures sufficient resolution and bandwidth to capture degradation or failure precursors.
5.	Sensor Integration Capability	FOCRAMS modules must support plug-and-play integration with at least 5 sensor types: accelerometer, thermocouple, dielectric, strain gauge, and FBG optical sensors.	Enables extensibility and flexibility in configuration across car subsystems.
6.	Edge Processing Capability	Onboard edge units must execute FFT, Kalman filter, and CNN/LSTM inference within 5 ms per frame.	Local analytics reduce telemetry load and enable sub-10 ms system response.
7.	Communications Interface	A. Primary data interface must support 1000BASE-T1 (Automotive Ethernet) and CAN FD (2 Mbps).	Supports high-throughput, reliable

		B. All links must be EMI shielded and real-time safe (deterministic latency ≤ 5 ms).	data delivery to the SECU and FIA logger.
8.	Power and Safety Compliance	A. System must operate at 9-16 VDC, with total draw < 10 W.	Compliance ensures safe, durable operation under race conditions.
		B. All circuits fused and shielded per FIA 2025 electrical safety guidelines.	
		C. Enclosures must meet IP67 and fire-retardant standards.	
9.	Fluid Diagnostics Performance	A. Oil condition module must measure viscosity, dielectric constant, and contaminant presence with update interval ≤ 1 s.	Accurate, timely detection of lubricant or hydraulic failure is critical in racing.
		B. Brake fluid sensor must detect water ingress and glycol breakdown within 1% accuracy.	
10.	Brake System Analytics	A. Pad strain and temp (via FBG) must be logged at 100 Hz, with accuracy of $<2\%$ strain and $\pm 1^\circ\text{C}$.	Supports real-time wear prediction and brake balance optimization.
		B. Disc thermal gradient must be sampled at 50 Hz, with mapped resolution of $\pm 5^\circ\text{C}$.	

II. Systems Modules

The FOCRAMS (Formula One Racing Car Components Reliability Assessment and Monitoring System) is composed of several specialized modules, each engineered to perform specific real-time monitoring and diagnostic tasks within the high-performance and harsh environment of Formula One racing. The following sections detail each module and its internal components, describing their respective functions and integration within the FOCRAMS architecture.

1. Fluid-Quality Monitoring Module

The Fluid-Quality Monitoring Module is responsible for assessing the condition of engine oil and brake fluid in real time. This module contains a MEMS viscometer to measure oil viscosity, a dielectric constant sensor to monitor fluid composition, and a conductivity sensor to detect water ingress and glycol breakdown in brake fluid. It also integrates pressure and temperature sensors for contextual analysis. These signals are processed through signal conditioning circuits and transmitted via a CAN or RS485 interface.

2. Brake-Pad and Brake-Disc Monitoring Module

The Brake-Pad Monitoring Module provides real-time measurements of brake pad force, wear, and localized temperature. Fiber-optic Fiber Bragg Grating (FBG) strain and temperature sensors are embedded within the brake pad friction material. These sensors are connected through rugged optical fiber connectors to an FBG interrogator, typically capable of reading 8 to 16 channels. The data collected is preprocessed and output through CAN or Ethernet, allowing engineers to analyze brake pad performance dynamically.

To monitor brake disc behavior, the Brake-Disc Sensor Module incorporates high-temperature K-type thermocouples and wireless MEMS temperature transmitters, or

alternatively, a slip-ring telemetry system. Power for these sensors can be supplied through onboard batteries or energy harvesters. In some configurations, strain gauges may also be mounted on the rotors to detect structural stress. Data from this module is collected by a receiver or gateway and sent to the central processing unit or ECU.

3. Phase Change Materials (PCM) Status Monitoring Module

The PCM-Status Monitoring Module tracks the state of the PCM materials. Distributed thermistors or fiber-optic temperature sensors are installed around the PCM zones. A dedicated signal processing unit detects key thermal behaviors, such as the melting point and plateau regions, allowing the system to infer thermal buffering capacity. Based on this data, FOCRAMS can execute thermal control strategies or modify brake bias and make adjustments to brake balance.

4. Sensor Node Network

The Sensor Node Network monitors general vehicle dynamics and conditions, including vibration, strain, pressure, and temperature. Each node includes a tri-axial high-G accelerometer with at least 10 kHz bandwidth, a high-precision temperature sensor accurate to ± 0.5 °C, and optional strain or pressure sensors. These nodes feature integrated microcontrollers for data digitization and preprocessing, with SPI or I²C communication and rugged IP67-rated enclosures for environmental protection.

5. Edge Processing Unit

The Edge Processing Unit executes local signal analytics, data fusion, and machine learning inference. It typically features a high-performance SoC or microcontroller such as an ARM Cortex-M7 or a Jetson Xavier NX module, running a real-time operating system or Linux. Embedded AI models such as Convolutional Neural Networks (CNNs) or Long Short-Term Memory (LSTM) are deployed for predictive diagnostics. The unit includes memory and storage for logging, Controller Area Network with Flexible Data-rate (CAN FD) or Automotive Ethernet interfaces, and onboard power regulation and watchdog systems for fault tolerance.

6. Communication Backbone

The Communication Backbone ensures seamless and reliable data flow between all FOCRAMS modules. It includes CAN FD transceivers, automotive-grade Ethernet PHYs (Physical Layers) (1000BASE-T1), EMI-shielded cabling, and safety-critical communication protocols such as Time-Sensitive Networking (TSN). A central junction box or gateway node arbitrates messages and maintains low-latency communication across the network.

7. Power and Housing Subsystem

The Power and Housing Subsystem supplies and protects all electronic components. It includes a regulated DC-DC power converter suitable for automotive voltage ranges, current fuses, Electromagnetic Interference (EMI) filters, and Electrostatic Discharge (ESD) protection circuitry. All modules are enclosed in IP68-rated polycarbonate or aluminum housings and connected using fire-retardant, high-temperature wiring. Isolation standoffs and mounting brackets provide mechanical stability and ease of installation.

Module Integration Overview

All FOCRAMS modules are engineered to work as an integrated, real-time data acquisition and analytics ecosystem. Each sensor node gathers high-fidelity signals from its respective environment—whether fluid channels, brake hardware, or thermal zones—and transmits this data via SPI/I²C to local microcontrollers. These controllers preprocess the information and forward relevant metrics to the edge processing unit through a unified communication backbone consisting of CAN FD or automotive Ethernet. The edge processor, equipped with real-time AI inference capabilities, fuses data from multiple domains (mechanical, thermal, and fluidic) to generate a comprehensive health assessment of the vehicle's components. The thermal-management and PCM-status modules interface closely, allowing automated adjustments of SMA-based airflow devices in response to thermal load or phase-change conditions. The power and housing subsystem ensures electrical stability and environmental protection for all modules, maintaining system integrity during high-speed operation. All critical outputs are routed to the central data logger or telemetry hub, ensuring compliance with FIA regulations and supporting strategic decision-making in real time. In essence, the FOCRAMS nodes will gather synchronous multi-domain data and perform on-board fusion. By fusing heterogeneous data at the edge, the system can detect subtle precursors to failure that might be missed by any single sensor.

Each FOCRAMS module is built with motorsport-grade and industrial components to withstand extreme conditions and provide precise, real-time analytics. Together, they deliver a full-spectrum reliability and thermal management platform, enabling Formula One teams to maintain component health, optimize strategy, and meet FIA compliance with cutting-edge predictive insight.

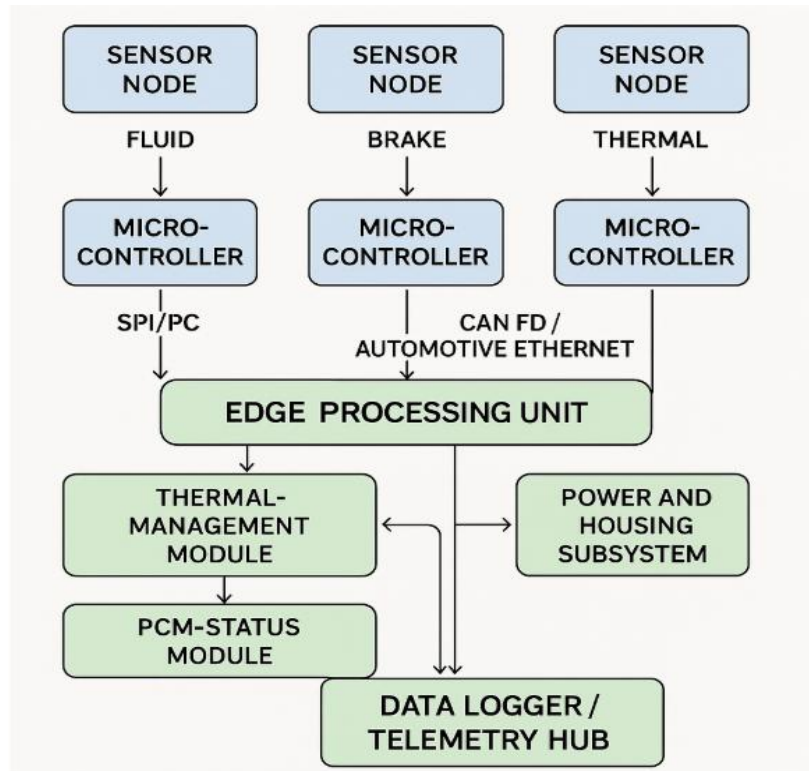


Figure 1 Flow chain of Data Acquired from Sensor Nodes to Processing Units

Development Schedule

FOCRAMS Development Schedule

Month 1: Project Kickoff & Requirements Finalization

Establish the development team, define technical and performance requirements, finalize the module architecture, and assign responsibilities. Begin procurement of critical long-lead components (e.g., FBG interrogators, edge computing boards).

Month 2: Design Schematics & CAD Prototyping

Complete electronic schematics, sensor layout, PCB designs, and initial CAD models for enclosures and PCM integration. Simulate sensor placement and heat loads.

Month 3: Sensor Calibration Setup & Software Framework

Set up calibration rigs for pressure, strain, temperature, and fluid sensors. Begin development of embedded software architecture including real-time operating systems and communication drivers.

Month 4: Module-Level Prototyping (Phase 1)

Fabricate early prototypes of sensor nodes, brake-pad sensing module, and fluid monitoring systems. Validate functionality of each sensor independently on bench setups.

Month 5: Edge Processing and AI Integration

Develop and deploy test CNN/LSTM models for vibration pattern recognition and oil degradation prediction. Integrate AI inference engine on edge units and perform stress testing.

Month 6: Mechanical Assembly and Housing Testing

Assemble modules with enclosures and conduct initial IP68 and vibration testing. Install SMA-based thermal flaps and verify actuation logic.

Month 7: System Integration Testing (Lab)

Interconnect all modules via CAN FD and Automotive Ethernet. Run full system simulation and log data under lab-controlled conditions.

Month 8: On-Vehicle Test Fit and Dry Runs

Install the FOCRAMS unit on a stationary F1 chassis or test mule. Perform dry runs (no track exposure) to validate installation and electrical compatibility.

Month 9: Initial Track Trials and Wind Tunnel Testing

Conduct on-track low-speed tests with real vehicle systems. Capture sensor logs, test telemetry latency, and validate cooling integration and thermal response. Simultaneously, carry out wind tunnel testing to ensure that the FOCRAMS unit, including all sensors, wiring, and enclosures, does not adversely affect aerodynamic flow or ground effect characteristics. Make necessary adjustments to mounting locations or casing shapes based on aerodynamic data.

Month 10: Data Analysis and Refinement

Refine algorithms using on-track and wind tunnel data. Update firmware and AI models for improved detection of anomalies, phase changes, and braking patterns. Ensure thermal and aerodynamic optimizations are implemented.

Month 11: Final Validation and Stress Testing

Execute full-performance testing at racing speeds. Run fault injection scenarios to validate warnings and redundancies. Certify reliability under FIA standards.

Month 12: Documentation and Deployment

Prepare technical documentation, final assembly guides, and operational manuals. Package FOCRAMS for race deployment and deliver final report.

This 12-month schedule ensures that FOCRAMS transitions from concept to a tested, operational system within one racing season, meeting the technical, regulatory, and environmental standards of Formula One.

Expense Estimation

Estimated FOCRAMS System Cost Table

Module	Description	Estimated Cost (USD)
Fluid-Quality Monitoring Module	MEMS viscometer, dielectric, conductivity sensors	\$300
Brake-Pad Sensing Module	FBG sensors, interrogator, optical connectors	\$6,000
Brake-Disc Sensor Module	Thermocouples, wireless MEMS, telemetry gateway	\$3,300
Thermal-Management Module	PCM inserts, thin-film sensors, SMA flap actuators	\$1,000
PCM-Status Monitoring Module	Thermistors/fiber sensors, signal processor	\$100
Sensor Node Network	Accelerometers, strain/pressure/temp sensors	\$800
Edge Processing Unit	Jetson Xavier NX or equivalent, AI support	\$700
Communication Backbone	CAN FD, Automotive Ethernet, harness, shielding	\$300
Power and Housing Subsystem	DC-DC supply, fuse, enclosures, wiring	\$300
Hardware Subtotal		\$12,800
Software & AI Development	Embedded firmware, ML models, system calibration	\$50,000
Integration and Testing	Assembly, validation, lab equipment, wind tunnel leasing	\$150,000
Labor Cost	Engineering and technician labor	\$20,000
Logistics and Shipping	Packaging, transport, and insurance	\$5,000
Spare Components and Redundancy	Backup sensors, critical hardware duplicates	\$5,000
Final Assembly and Installation	Vehicle integration and fitment	\$5,000
Total Estimated FOCRAMS Cost		\$247,800

Note: These costs are based on a single FOCRAMS kit intended for one Formula One car, using a mix of commercial off-the-shelf components and prototype-level development, inclusive of testing, labor, logistics, and redundancy planning.

AI and LLM Implications

The development of the FOCRAMS proposal has been significantly supported by the use of advanced AI tools, particularly large language models such as ChatGPT. These tools enabled me to explore and understand complex engineering concepts and principles underlying some of the most sophisticated technologies in Formula One. I utilized ChatGPT and similar LLMs to conduct in-depth research into component reliability assessment, sensor fusion methodologies, and real-time telemetry systems, all of which are critical in the motorsport context. Additionally, these tools proved invaluable for identifying and analyzing relevant patents, academic papers, and recent advancements in the field, both domestically and internationally. Through their assistance, I was able to consolidate a large volume of technical data into a coherent and innovative system architecture tailored for high-performance racing applications.

References:

- [1] L. Belkovic and I. Takács, "The Digitalization of Formula 1: Innovations from Car Design to Race Strategy," 2023 IEEE 21st Jubilee International Symposium on Intelligent Systems and Informatics (SISY), Pula, Croatia, 2023, pp. 000565-000570, doi: 10.1109/SISY60376.2023.10417955.
- [2] A. Gu, A. Andreyev, M. Terada, T. Rodgers and V. Viswanathan, "A Deep Learning Reconstruction Technique and Workflow to Enhance 3D X-ray Imaging Resolution and Speed for Electronics Package Failure Analysis," 2023 International Conference on Electronics Packaging (ICEP), Kumamoto, Japan, 2023, pp. 69-70, doi: 10.23919/ICEP58572.2023.10129701.
- [3] P. Karle, F. Fent, S. Huch, F. Sauerbeck and M. Lienkamp, "Multi-Modal Sensor Fusion and Object Tracking for Autonomous Racing," in IEEE Transactions on Intelligent Vehicles, vol. 8, no. 7, pp. 3871-3883, July 2023, doi: 10.1109/TIV.2023.3271624.
- [4] Kullu, O.; Cinar, E. A Deep-Learning-Based Multi-Modal Sensor Fusion Approach for Detection of Equipment Faults. *Machines* 2022, 10, 1105.
<https://doi.org/10.3390/machines10111105>
- [5] Qin Tang, Jing Liang, Fangqi Zhu, A comparative review on multi-modal sensors fusion based on deep learning, *Signal Processing*, Volume 213, 2023, 109165, ISSN 0165-1684, <https://doi.org/10.1016/j.sigpro.2023.109165>.
- [6] Buabeng, Albert & Simons, Anthony & Frempong, Nana Kena & Yevenyo Ziggah, Yao. (2022). Predictive Maintenance Model Based on Multisensor Data Fusion of Hybrid Fuzzy Rough Set Theory Feature Selection and Stacked Ensemble for Fault Classification. *Mathematical Problems in Engineering*. 2022. 10.1155/2022/4372567.
- [7] Md. Abdur Rahim, Md. Mustafizur Rahman, Md. Shofiqul Islam, Abu Jafar Md. Muzahid, Md. Arafatur Rahman, Devarajan Ramasamy, Deep learning-based vehicular engine health monitoring system utilising a hybrid convolutional neural network/bidirectional gated recurrent unit, *Expert Systems with Applications*, Volume 257, 2024, 125080, ISSN 0957-4174, <https://doi.org/10.1016/j.eswa.2024.125080>.
- [8] Khanmohamadi, M., & Guerrieri, M. (2024). Advanced Sensor Technologies in CAVs for Traditional and Smart Road Condition Monitoring: A Review. *Sustainability*, 16(19), 8336. <https://doi.org/10.3390/su16198336>

[9] Gültekin, Ö., Cinar, E., Özkan, K., & Yazıcı, A. (2022). Real-Time Fault Detection and Condition Monitoring for Industrial Autonomous Transfer Vehicles Utilizing Edge Artificial Intelligence. *Sensors*, 22(9), 3208. <https://doi.org/10.3390/s22093208>

[10] Kociszewski, R., & Wojtkowski, W. (2025). Remote Vibration Monitoring of Combustion Engines Utilising Edge Computing. *Electronics*, 14(11), 2118. <https://doi.org/10.3390/electronics14112118>

[11] Karabacak, Y. E. (2024). Condition monitoring of internal combustion engines with vibration signals and fault detection by using machine learning techniques. *International Journal of Automotive Engineering and Technologies*. <https://doi.org/10.18245/ijaet.1251886>